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Calculus/Tables of Derivatives

Wikipedia has related information at ***Differentiation rules*** (https://en.wikipedia.org/wiki/Differentiation_rules).

General Rules

$$\frac{\mathrm{d}}{\mathrm{d}x}(f+g)=\frac{\mathrm{d}f}{\mathrm{d}x}+\frac{\mathrm{d}g}{\mathrm{d}x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(c\cdot f)=c\cdot\frac{\mathrm{d}f}{\mathrm{d}x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(f\cdot g)=f\cdot\frac{\mathrm{d}g}{\mathrm{d}x}+g\cdot\frac{\mathrm{d}f}{\mathrm{d}x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{f}{g}\right)=\frac{-f\cdot\frac{\mathrm{d}g}{dx}+g\cdot\frac{\mathrm{d}f}{\mathrm{d}x}}{g^2}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}[f(g(x))]=\frac{\mathrm{d}f}{\mathrm{d}g}\cdot\frac{\mathrm{d}g}{\mathrm{d}x}=f'(g(x))\cdot g'(x)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}f(x)g(x)=\sum_{i=0}^{n}\binom{n}{i}f^{(n-i)}(x)g^{(i)}(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{f}\right)=-\frac{f'}{f^2}$$

Powers and Polynomials

- $\frac{\mathrm{d}}{\mathrm{d}x}(c)=0$
- $\frac{\mathrm{d}}{\mathrm{d}x}x=1$
- $\frac{\mathrm{d}}{\mathrm{d}x}x^n=nx^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x}=\frac{1}{2\sqrt{x}}$
- $\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{x}=-\frac{1}{x^2}$
- $\frac{\mathrm{d}}{\mathrm{d}x}(c_nx^n+c_{n-1}x^{n-1}+c_{n-2}x^{n-2}+\cdots+c_2x^2+c_1x+c_0)=nc_nx^{n-1}+(n-1)c_{n-1}x^{n-2}+(n-2)c_{n-2}x^{n-3}+\cdots+2c_2x+c_1$

Trigonometric Functions

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x)=\cos(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\sin(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan(x) = \sec^2(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\cot(x) = -\csc^2(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\sec(x) = \sec(x)\tan(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\csc(x) = -\csc(x)\cot(x)$$

Exponential and Logarithmic Functions

- $\frac{\mathrm{d}}{\mathrm{d}x}e^x = e^x$
- $\frac{\mathrm{d}}{\mathrm{d}x}a^x = a^x\ln(a) \qquad \text{if } a > 0$
- $\frac{\mathrm{d}}{\mathrm{d}x}\ln(x) = \frac{1}{x}$
- $\frac{\mathrm{d}}{\mathrm{d}x}\log_a(x) = \frac{1}{x\ln(a)} \qquad \text{if } a > 0\,,\ a \neq 1$
- $\frac{\mathrm{d}}{\mathrm{d}x}(f^g) = \frac{\mathrm{d}}{\mathrm{d}x}\left(e^{g\ln(f)}\right) = f^g\left(f'\frac{g}{f} + g'\ln(f)\right)\ , \qquad f > 0$
- $\frac{\mathrm{d}}{\mathrm{d}x}(c^f) = \frac{\mathrm{d}}{\mathrm{d}x}\left(e^{f\ln(c)}\right) = c^f\ln(c)\cdot f'$

Inverse Trigonometric Functions

$$\frac{\mathrm{d}}{\mathrm{d}x}\arcsin(x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\arccos(x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\arctan(x) = \frac{1}{x^2+1}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arccot}(x) = -\frac{1}{x^2+1}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcsec}(x) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arccsc}(x) = -\frac{1}{|x|\sqrt{x^2-1}}$$

Hyperbolic and Inverse Hyperbolic Functions

$$\frac{\mathrm{d}}{\mathrm{d}x}\sinh(x)=\cosh(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\cosh(x)=\sinh(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\tanh(x)=\operatorname{sech}^2(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{sech}(x)=-\tanh(x)\operatorname{sech}(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\coth(x)=-\operatorname{csch}^2(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{csch}(x)=-\coth(x)\operatorname{csch}(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arsinh}(x)=\frac{1}{\sqrt{x^2+1}}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcosh}(x)=\frac{1}{\sqrt{x^2-1}}\ ,\ x>1$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{artanh}(x)=\frac{1}{1-x^2}\ ,\ |x|<1$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arsech}(x)=-\frac{1}{x\sqrt{1-x^2}}\ ,\ 0<x<1$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcoth}(x)=\frac{1}{1-x^2}\ ,\ |x|>1$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcsch}(x)=-\frac{1}{|x|\sqrt{1+x^2}}\ ,\ x\neq 0$$

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